

Confidential

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Polybee Pte Ltd (“Polybee”)

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Investment Memo

Company Profile

Polybee has developed a differentiated drone-based pollination system delivering up to three times the efficacy of natural ventilation, supported by AI-driven yield forecasting proven to deliver 17–30% yield uplifts in trials. The business model charges per hectare, per crop cycle, generating rapid ROI for growers and gross margins of up to 65%. Strategic partnerships with Bayer Crop Science and the Western Growers Association provide strong market access.

Investment Recommendation

Proceed with an investment of US 1.0 m from PIIF 6 into Polybee’s US 2.5 m Seed Round, subject to Investment Committee approval, to acquire 5.73% at a US 15.0 m pre-money valuation.

Company Name	Polybee	Proposed security	Preference Shares
Year Founded	2019	Round Stage	Seed
Headquarters	Singapore	PIIF 6 Investment	US 1.0m
Sector	AgTech	Target PIIF %	5.73%
Technology	Yield Forecasting	Round Proceeds	US 2.5m
Founder(s)	Sidd Jadhav (CEO)	ESOP Expansion?	No
NTM Rev	US 2.1m	Pre-Money Valuation	US 15m
Pipeline 2025/6	US 4.0m	Seed Key Milestone(s)	Capture NTM Rev
No. of Staff	9	Series A+ TRL	9
Monthly Burn	US 100 k	PPS	US 10.186
Cur. monthly runway	5.2 months	Equity Sold	27.03%
Current TRL	9		

Investment Recommendation

Investment Summary	Polybee is a Singapore-based agtech company offering drone-based pollination and AI-powered yield forecasting. The company's Aerodynamically Controlled Pollination (ACP) delivers up to three times the pollination efficacy of natural ventilation in greenhouse strawberries. Yield forecasting trials with major growers have shown 17–30% uplift in spinach yields. The business model charges per hectare, per crop cycle, providing rapid ROI for growers and supporting 65% gross margins in real-world deployments.
Company & Market	Founded in 2019 as an NUS spin-out, Polybee operates at TRL 8–9 with a team of nine and advisors from Bayer, Monsanto, and Blue River Technology. Target markets include high-value crops in the U.S., Australia, New Zealand, and Asia. The addressable market for pollination, yield forecasting, and stress detection is estimated at US 5.2b, with a U.S. serviceable obtainable market of US 18m for priority crops. Macro tailwinds include declining natural pollinators, climate variability, water scarcity, and labour shortages.
Investment Thesis & Risks	The investment case rests on Polybee's strong early traction with Tier 1 growers and distributors, differentiated technology applicable to both open-field and greenhouse crops, and a capital-light service model with high gross margins. Key risks include reliance on converting Stage 1 trials into full-scale Stage 2 contracts, dependency on DJI drone supply in light of U.S. regulatory scrutiny, and a short runway if scaling ahead of revenue conversion. Mitigations include alternative hardware pathways, staged sales processes, and a large qualified pipeline.
Financials & Valuation	Pre-money valuation is US 15 m, targeting a US 2.5m Seed round. Historical revenue fell from SG 322k in 2022 to SG 86k in 2024 due to a pivot from seed-industry pilots to direct-to-grower sales. The current 2025–2026 pipeline is US 2.1m, mostly in Stage 1 trials. Gross margins from mature deployments are estimated at 65%. Monthly burn is approximately SG 97k with a 5.1-month runway, extendable to 15–18 months post-raise. Series A is targeted for late 2026 at a US 40–50m pre-money.
Investment Terms	Seed Preference Shares. Target round size: US 2.5m (PIIF6 allocation: US 1m). Liquidation preference: Non-participating, 1x Board: One seat for lead investor and/or option to appoint an observer Anti-dilution: Broad-based weighted average. Reserved matters: Customary, with additional protections for lead investor. The recommendation is to proceed with the investment, subject to investment committee approval.

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Conflict of Interest Disclosure

Paspalis Capital discloses a potential conflict of interest. A condition of the proposed investment includes the delivery of a fully-executed serviced office agreement with Darwin Innovation Hub Venues Pty Ltd (DIHV), a related entity of Paspalis Capital. Harley Paroulakis is the sole director of both. The ownership structures are different, but the ultimate owners are related.

The Darwin Innovation Hub (DIH) has offered Polybee Australia Pty Ltd an 18-month serviced office agreement. The agreement will renew automatically unless either party gives at least 90 days' written notice before the end date. Either party may also end the agreement with 90 days' notice at any time.

To suit the company's needs, DIH has provided a range of office suite options from 12.5m² to 35.9m², with monthly licence fees between AUD 1,500 for a 1-2 person office with no window) and AUD 4,400 (larger offices) plus GST. All options are offered at retail market rates, as detailed in the attached document.

Commercial Rationale

Polybee's seed round supports its planned expansion into Northern Australia. The decision to establish a presence in Darwin aligns with its focus on high-value crops, including mangoes, melons, and citrus.

Polybee has visited the DIH and will utilise the DIH's communal area. The Company has informed us that an MOU is to be signed with NT Plant Industries. The company's presence at the DIH positions it to engage with these customers and R&D partners, supporting yield improvement across high-value mango, melon, and citrus crops in the Northern Territory. These projects include:

1. Working with Red Dirt Farms, a melon grower in the NT, and NT Plant Industries to trial drone-based imaging and 3D fruit modelling for early stress detection and precise harvest forecasting.
2. Polybee is also expected to partner with Razor Rock Farms, Fortuna Produce, and NT Plant Industries to adapt its 3D tracking and thermal models for accurate mango yield and size prediction.

DIH is a Grade A commercial office and co-working space in Darwin's CBD. A local presence enables Paspalis to support and monitor its investment as the company expands its market in the Northern Territory and across Australia. The DIH also hosts long-term tenants such as the NT Plant Industries, which works closely with horticulture companies and organisations operating in urban remote areas around Darwin, Northern Territory. These connections may help the company identify new sales opportunities.

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About Polybee

Founded in 2019, Polybee is a Singapore-based agritech startup that creates autonomous micro-drones. These drones use AI, computer vision, and smart automation to perform precise pollination and yield forecasting. This means that growers can boost productivity in indoor farms with less labour and more consistency.

The company was founded as a spin-off from the National University of Singapore and officially incorporated in 2019, led by CEO Siddharth Jadhav. It has raised ~US 4m from investors including Elev8, SEEDS Capital, and corporate partners, including the Temasek Foundation.

Polybee's main innovation is aerodynamically controlled pollination. The drones generate targeted airflow to shake pollen loose without touching the flowers, mimicking bees while reducing the risk of damage and disease. Onboard 3D cameras and multispectral sensors collect plant data such as fruit size, stem thickness, stress levels, and estimated weight. AI models convert this data into yield forecasts and actionable insights for growers.

The company offers an evidence-based pricing model, charging between US500-1000 per Ha per crop cycle. It has tested its technology with overseas growers, including in the UK for strawberry pollination, and has recently expanded into Australia by establishing an ANZ headquarters in Melbourne to serve regional growers and advance R&D.

Corporate Structure

Polybee Pte Ltd

Polybee Pte Ltd is an exempt private company limited by shares, incorporated in Singapore on 13 September 2019 under the Companies Act 1967. The company is recorded as a live entity. It has a single class of ordinary shares, with an issued and paid-up share capital of SGD 1,000,000 comprising 1,000,000 ordinary shares. All shares are held by founder and CEO Siddharth Sunil Jadhav, an Indian national and Singapore resident, giving him full ownership and control. Dennis Poh Wei Song, a Singapore citizen, serves as the company secretary. From a compliance and governance perspective, the company holds a Certificate of Good Standing from ACRA dated 9 August 2025, confirming it is properly incorporated, active, and in compliance with its statutory obligations.

Polybee Australia Pty Ltd

Polybee Australia Pty Ltd (ACN 65 683 205 978) presents a straightforward corporate and compliance profile, with no immediate red flags. Its wholly owned structure ensures full strategic and operational control by the Singapore-incorporated parent, Polybee Pte Ltd, aligning the subsidiary's activities with group objectives. The single-shareholder (Polybee Pte Ltd), dual-director arrangement simplifies governance with the nominee director Riten Patel providing assurance of compliance with the Australian-resident director requirement. From a compliance perspective, the company holds all required Australian registrations, including ACN, ABN, TFN, and GST registration, with appropriate quarterly reporting arrangements. Overall, the subsidiary's legal and governance framework is sound, but given its early stage and tight control, ongoing monitoring of governance practices, director performance, and compliance milestones will be important for investors.

Team Maturity & Staffing

Polybee's leadership team combines deep technical expertise in robotics, AI, and agricultural automation with extensive agribusiness and global seed industry experience. CEO and founder Siddharth Jadhav brings a strong R&D background from NUS and seven years of product development in ag robotics, while VP of Commercial Operations Pankaj Malik adds over two decades in senior roles at Bayer, DuPont, and Monsanto. VP of Engineering Samir Otiv, a former founder and IIT graduate, has scaled industrial automation platforms globally. Strategic advisors include John Purcell, ex-Bayer and Monsanto R&D leader with 35 years in vegetable crop innovation, and Jorge Heraud, co-founder of Blue River Technology, acquired by John Deere for US 305m, with extensive experience in ag robotics and autonomy.

The current board structure lists Jadhav as the sole director, which centralises control; as part of the priced seed round, appointing a Paspalis representative could strengthen governance, oversight, and strategic alignment.

Sidd Jadhav, CEO

Siddharth Jadhav is the founder and CEO of Polybee, an agritech company that utilises drone technology to enhance pollination and yield forecasting in controlled-environment agriculture. He holds a background as a research scientist at the National University of Singapore (NUS), where he worked on drone dynamics, control systems, and automation. That experience helped set the stage for launching Polybee in 2019 as a NUS spin-off venture.

He began his career as a research scientist at the National University of Singapore, working on drone dynamics, control systems, and automation. He holds a degree in chemical engineering from the Birla Institute of Technology and Science, Pilani, and moved into entrepreneurship through the Lean LaunchPad programme.

Siddharth has more than seven years of experience leading product development in robotics, AI, and automation for agriculture. At Polybee, he drives innovation in autonomous micro-drone systems that use targeted airflow to pollinate plants without contact, reducing the risk of damage and disease. These drones also gather plant data for yield forecasting through computer vision and AI.

Under his leadership, Polybee has expanded internationally, including the opening of its ANZ headquarters in Melbourne to advance research and partnerships with growers in the region.

Pankaj Malik, VP Operations (Ex Bayer, Monsanto, Du Pont)

Pankaj Malik has over two decades of leadership experience in the global agribusiness sector, specialising in seeds, crop science, and technology-driven market growth. He joined Polybee in January 2023 to lead commercial operations, with responsibility for scaling yield estimation and autonomous pollination solutions in Australia, New Zealand, Europe, and North America.

Prior to Polybee, Pankaj was Regional Director for South East Asia at Enza Zaden, overseeing marketing, sales, R&D, supply chain, and finance functions across multiple countries. He previously held senior positions at Bayer Crop Science, including Head of Seed Business for Indonesia and Malaysia, and Head of Marketing for rice seeds and traits in India. His earlier career includes commercial leadership roles at DuPont and Monsanto.

Pankaj is also a Council Member of the World Agriculture Forum, contributing to global policy and innovation initiatives for sustainable agriculture. His career reflects a consistent ability to deliver commercial growth by aligning advanced agricultural technologies with market needs.

Samir Otiv, VP Engineering (Ex-Founder, Ex IIT)

Samir Otiv joined Polybee in May 2025 as Vice President of Engineering, leading the development and scaling of the company's autonomous pollination and yield forecasting technologies for the global fresh produce sector. His role focuses on delivering high-impact agricultural automation solutions that increase yields, reduce waste, and improve operational efficiency for growers in Australia, New Zealand, the United States, and other key markets.

Before Polybee, Samir spent nearly a decade as Chief Technology Officer at Maximl, a venture-backed industrial technology startup providing intelligent field automation for deskless workers in heavy industries such as oil and gas and chemicals. He oversaw product strategy, engineering execution, and platform development, building scalable solutions for collaborative project management, connected worker systems, and sustainability initiatives.

Earlier in his career, Samir held research internships at Intel Corporation and Tata Research Development and Design Centre, working in CPU architecture optimisation and applied research. He holds a Dual Degree (BTech & MTech) in Computer Science and Engineering from the Indian Institute of Technology, Madras. His background in industrial automation, AI-driven platforms, and scaling complex technology products positions him to drive Polybee's engineering and product innovation agenda.

Polybee Team

Name	Role	Role Type	Region
Siddharth Jadhav	CEO	Exec	Singapore
TBC	Sr. CV Engineer - OF	Engineering	Singapore
Sarthak	Sr. Robotics Engineer	Engineering	Singapore
Majid	Sr. CV Engineer - OF	Engineering	Singapore
Damodar	Sr. CV Engineer - GH	Engineering	Singapore
Pankaj	VP - commercial	Exec	Singapore
Samir	VP - engineering	Exec	Singapore
Tzi Wah	Ops associate	Operations	Singapore
Shobhit	Full-stack Developer	Engineering	India

John Purcell (Ex Bayer Crop Science, Monsanto)

John Purcell is a senior agricultural technology leader with over 35 years of experience in R&D, commercialisation, and executive management in the global seed and fresh produce sectors. He is the Founder and President of AgTech Advisory Services where he provides strategic guidance to companies advancing agricultural innovation.

From 2020 to 2023, John was CEO and later Chief Strategy Officer of Unfold, a venture-backed vertical farming seed and digital solutions company. Under his leadership, Unfold developed crop varieties and technology tailored to controlled-environment agriculture, forging partnerships with major indoor farming operators.

Prior to Unfold, John served as Head of Vegetables R&D at Bayer Crop Science, overseeing global research, breeding, and product development. He spent nearly three decades at Monsanto in senior technology and commercial roles, including Vice President and Global R&D Lead for Monsanto Vegetables, where he drove advancements in seed genetics, trait development, and market adoption across diverse horticultural crops.

John began his career as a Postdoctoral Researcher at the USDA after earning a Ph.D. in Molecular and Cellular Biology from the University of Massachusetts Amherst, specialising in insect biochemistry, and a B.S. in Chemistry from Washington and Lee University. His deep expertise in vegetable crop R&D, combined with leadership across both corporate and startup environments, makes him a valuable strategic resource for companies like Polybee seeking to scale disruptive agricultural technologies.

Jorge Heraud (Ex John Deere, IBM, Trimble, Stanford University, Co-Founder of an Exited Startup for US 305m)

Jorge Heraud is a technology entrepreneur and senior executive with more than three decades of experience in agricultural automation, robotics, and precision farming systems. He is currently CEO of TerraBlaster, and holds multiple board and advisory roles with leading agtech companies, including Chairman of Dogtooth Technologies and Board Member of GPR, RootWave, and Agtonomy. Through his consulting firm, Heraud Consulting LLC, he advises startups and corporates on scaling autonomous systems and precision agriculture solutions.

From 2011 to 2024, Jorge was CEO and Co-founder of Blue River Technology, where he led the development of advanced computer vision and machine learning platforms for precision weed control. The company's "see and spray" technology reduced herbicide use by up to 90% and delivered significant environmental and economic benefits for growers. Blue River was acquired by Deere & Company in 2017 for US 305m¹, where Jorge went on to serve as VP of Automation & Autonomy, overseeing the integration and scaling of smart machine technologies across agriculture and construction.

Earlier in his career, Jorge spent over a decade at Trimble in engineering, business development, and leadership roles, including Business Unit Manager for the Agriculture Division. He has also held engineering positions at Digita SA and IBM. Jorge holds master's degrees in Electrical Engineering, Engineering Management, and Management (Sloan Fellow) from Stanford University, with a focus on entrepreneurship, marketing, and finance.

¹ <https://www.cnn.com/2017/09/06/deere-is-acquiring-blue-river-technology-for-305-million.html>

His track record of founding, scaling, and exiting a world-leading agtech robotics company, combined with deep industry and technology expertise, makes him a highly strategic figure for driving innovation and market adoption in autonomous agricultural solutions.

Board Structure

According to records from the Accounting and Corporate Regulatory Authority (ACRA), Siddharth Jadhav, CEO, is currently the sole director of Polybee Pte Ltd. This structure centralises formal governance authority with the founder and may limit independent oversight and investor influence in strategic decision-making.

As part of the current priced seed round, it should be considered whether Paspalis Capital Partners should negotiate the right to appoint a representative to the board of directors. Such an appointment would provide direct visibility into company operations, enable more active participation in strategic direction, and help safeguard investor interests.

Market Size & Tailwinds

Market Size

Market Size

Polybee estimates that its pollination, yield forecasting and stress detection technology suite can address a total addressable market opportunity (TAM) of US 5.2b. This is based on 27 crops and their total production area, and expected value generated to farmers per hectare of between US 300-400 per Ha.

To estimate the serviceable obtainable market (SOM) opportunity for Polybee, we constrained the above market size methodology to crops in the United States where Polybee has pilots underway (blueberries, kiwi fruit, strawberries, and tomatoes) and those where clear value savings have been proven (spinach and broccoli). These crops are also significant to the United States. Over 20% of the area of production for these crops is in the United States, and a majority of U.S vegetable and fruit farms within Polybee's target crop list suggest larger farms (500 Ha+) overwhelmingly contribute to production. Based on this information, we then applied a 10% penetration rate to understand the realistic potential of a recurring revenue opportunity for Polybee.

		Hectares (Ha)	US m		
Crop	Status	USA	TAM	SAM	SOM
Blueberries	Pilot	51,315	165	51	5
Broccoli	Value proven	62,605	147	31	3
Kiwi fruit	Pilot	1,943	71	2	0
Spinach	Value proven	29,542	57	15	1
Strawberries	Pilot	22,986	166	23	2
Tomatoes	Pilot	122,135	371	61	6
		290,526	976	183	18

Source: Polybee, Various Online Sources

We estimate SOM to be US 18m based on U.S entry alone and Polybee's target crops.

SOM figures are highly sensitive to the farmgate value per hectare of the crop which influence higher value capture charged to growers. Higher value crops such as blueberries and strawberries are expected to generate value of US1k per Ha, than tomatoes at US 500 per Ha. This SOM forecast is highly sensitive to the adoption of Polybee's solution in a competitive market, particularly with larger growers. The leveraging of Polybee's Tier 1 distribution partners such as Bayer and DJI and a

scalable hardware-light business model are contributing factors which could see higher penetration rates in these U.S crops alone.

Pollination Market Size

For 2024, the US Department of Agriculture and the National Agricultural Statistics Service (NASS) estimated that producers spent more than US 400m on pollination services in the United States alone. The value of providing pollination services has risen such that beekeepers have reported making more from these activities than from honey production since 2022.²

Importantly, almonds accounts for 83% of the total pollination services market in the United States. Most of all US almonds are grown in California, and it has been estimated that 80% of all honeybees in the US are transported to California every spring to pollinate almond flowers. If Polybee's artificial pollination technology can offer a compelling value proposition in the US pollination service market, there is significant service addressable market opportunity worth over US 167m per year.

Pollination Services Market in the U.S

Region	Acres	Value in US m
Pacific Northwest	240,600	18.3
Pacific Southwest	1,191,000	352.8
Midwest and Mountain	4,900	0.7
Northeast	183,800	18.5
South	46,700	4.1
Southeast	61,300	6.3

Source: USDA³

Some evidence exists to support the potential of Polybee's drone-based pollination solution.

In the US, there are approximately 115,000-125,000 beekeepers, most of whom are hobbyists, but 1,600 are commercial beekeepers and responsible for 60% of honey production.⁴ Most of these beekeepers supplement revenue from honey production with pollination services, transporting honeybees around the country to pollinate orchards and fields during pollination windows.⁵ It has been estimated that 75% of farmers are using smart hives, sensors, and hive-grade techniques to enhance natural pollinator management, while around 25% are using technology, such as drones, to replace natural pollinators.

Managers managed pollinators face similar threats to the wild pollinators: pesticides, crop monocultures, and Colony Collapse Disorder negatively affect bee health, sometimes killing them outright. This suggests a strong tailwind, which could see artificial pollination technology take adoption beyond 25% over the next 5-10 years.

² <https://www.ers.usda.gov/data-products/charts-of-note/chart-detail?chartId=112782#:~:text=For%202024%2C%20USDA%2C%20National%20Agricultural,from%20honey%20production%20since%202022.>

³ <https://www.ers.usda.gov/data-products/charts-of-note/chart-detail?chartId=112782#:~:text=For%202024%2C%20USDA%2C%20National%20Agricultural,from%20honey%20production%20since%202022.>

⁴ <https://www.agmrc.org/commodities-products/livestock-dairy-poultry/bees-profile>

⁵ <https://www.refood.co/stories/pollination-tech>

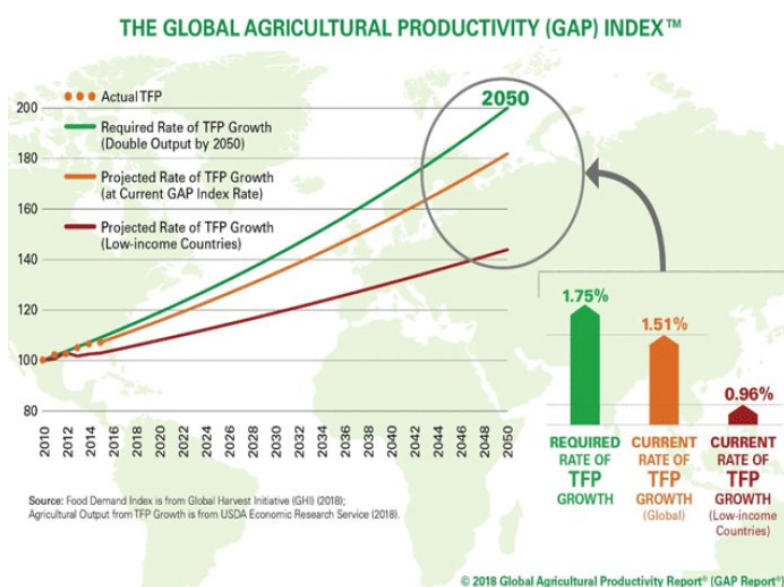
Market Tailwinds

Climate Variability is Making Yield Forecasting More Uncertain

Climate variability is making growing conditions less predictable. More frequent extreme weather events and seasonal shifts increase the value of early, accurate yield forecasts to adapt planting, irrigation, and harvest strategies.⁶ For example, rainfall and temperature that farmers used to count on are now highly erratic, complicating decisions on when to plant and harvest. Evidence indicates that weather patterns are indeed getting less reliable, with greater swings in precipitation and temperature from the new normal.⁷ Polybee's yield forecasting and stress detection give growers early, site-specific insights into crop health and growth patterns, allowing them to respond quickly to unexpected weather impacts. This means farmers can adjust management decisions in real time to protect yields and reduce losses in increasingly unpredictable conditions.

Global Food Demand-Supply Gap Is Expected to Grow

Overall demand is expected to outpace production in many regions. For instance, in a group of 76 developing countries studied by USDA, food demand is projected to grow about 2.6% annually from 2020 to 2030, faster than the 2.0% annual growth in food output.⁸ These trends underscore the need for significant gains in agricultural productivity to meet the world's rising food requirements. Polybee's artificial pollination, yield forecasting, and stress detection help close the food demand–supply gap by boosting crop yields, reducing losses, and improving resource efficiency.



⁶ <https://www.evokeag.com/climate-changes-effects-on-farming-and-agriculture/#~:text=No%20longer%20tomorrow%E2%80%99s%20problem%2C%20climate, scale%20commercial%20operations>

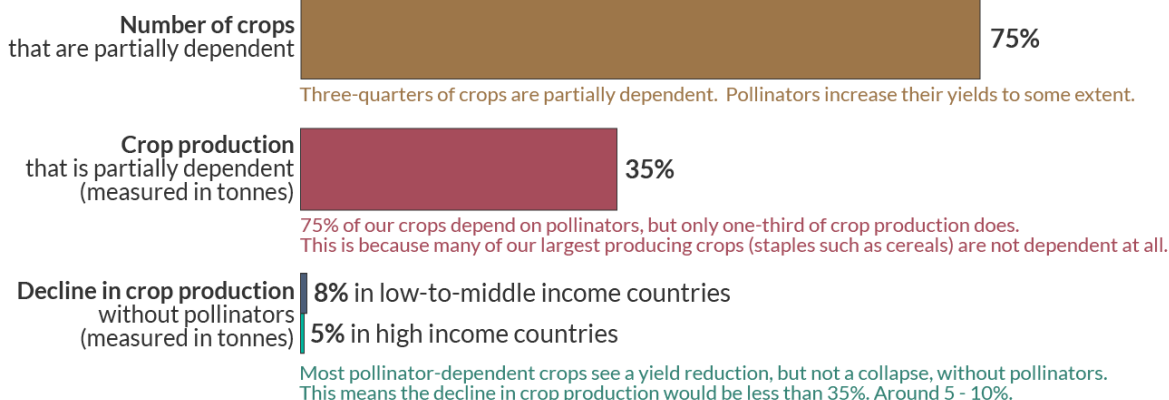
⁷ As above

⁸ <http://www.ers.usda.gov/amber-waves/2021/february/international-food-security-to-improve-by-2030-covid-19-related-income-declines-in-2020-temper-overall-improvement>

Decline in Natural Pollinators, Yet Needed by a Third of Plants

For a flower to produce a seed (and the fruit that surrounds it), it usually needs to mate with another flower. Plants have evolved to attract pollinators such as insects, who come for sweet nectar and leave loaded with pollen that they deposit on the next flower they visit. An estimated 35% of food crops require animal assistance for pollination.⁹ One analysis showed that more than 50% of native bee species in North America and Hawaii are declining, and another 24% are in serious peril.¹⁰ Pollinator decline is a serious threat to our food system: it could contribute to 5-10% or greater declines in food production, especially of high-income crops like coffee, cocoa, fruits, and nuts.¹¹

How much of global food production depends on pollinators?

**Our World
in Data**


Sources: Marcelo Aizen et al. (2019) and Alexandra-Maria Klein et al. (2006).
OurWorldinData.org – Research and data to make progress against the world's largest problems. Licensed under CC-BY by the author Hannah Ritchie.

Labour Shortages Risks On-Farm Productivity and Supply Targets

Agricultural labor shortages (“grower staff” shortages) are an increasingly acute problem worldwide. Many farmers are aging – the average farmer is over 55 in many regions – and not enough young workers are entering agriculture to replace retirees.¹² Farmworker roles are actually among the fastest-growing job categories; the World Economic Forum estimates agriculture will need 35m additional farm workers globally by 2030. In countries like the United States and Canada, the labor gap is already evident – over half of U.S. farmers have recently reported unfilled farm jobs and worker shortages.¹³ Polybee’s drone-in-a-box system, which self-flies and self-charges, reduces growers’ reliance on scarce labour by automating pollination and crop monitoring tasks. This means that farms can maintain productivity and consistency even when they cannot fill critical field roles.

⁹ <https://www.nifa.usda.gov/about-nifa/blogs/protecting-pollinators-critical-food-production>

¹⁰ https://www.biologicaldiversity.org/campaigns/native_pollinators/pdfs/Pollinators_in_Peril.pdf

¹¹ <https://ourworldindata.org/pollinator-dependence>

¹² <https://www.weforum.org/stories/2025/05/world-urgently-needs-more-young-farmers/#:~:text=young%20farmers%20into%20the%20field>

¹³ <https://www.fticonsulting.com/insights/articles/us-agriculture-navigating-labor-challenges-finding-solution#:~:text=Agriculture%20and%20food%20manufacturing%20are,5>

Drone Adoption and Risks

Potential DJI Ban

Polybee's solution relies on drone adoption and represents a capital cost to growers. Adoption also demands change of behaviour and culture of growers which have traditionally resisted scientific precision agriculture practices. Drone adoption in Australian agriculture is still maturing. In 2020, about 10% of Australian farms reported using drones.¹⁴ In the United States, drone adoption is considered to be low, but this is expected to change as regulations on drone crop-spraying ease.¹⁵ Starting 10 June 2024, the Federal Aviation Authority (FAA) streamlined the Part 137 UAS (unmanned aircraft system) certification process for agricultural operators.

Polybee uses DJI, a Chinese drone-manufacturer used and promoted by Polybee. DJI's dominance in the drone sector is well-documented, with some estimates placing its U.S. market share as high as 85%.¹⁶ This presents an opportunity for Polybee leverage DJI as a trusted brand within the U.S grower community.

Polybee's drone partner, DJI, has also come under scrutiny by U.S regulatory bodies in recent years, with the government taking steps to limit DJI's presence. The FY25 National Defence Authorisation Act (NDAA) mandates a formal security audit of DJI platforms by the end of 2025. If the DJI audit is not complete or DJI fails to pass, it will automatically be placed on the Federal Communications Commissions "Covered List", grounding new DJI products by denying them access to FCC-regulated bandwidths.¹⁷

Media outlet DroneDJ reports that of late June 2025, DJI's official U.S. online store is largely empty.¹⁸ According to a recent article in The Verge¹⁹, major retailers such as Best Buy, Amazon, B&H Photo, and Adorama report near-total depletion of DJI drone stock. These retailers have denied any official exit by DJI, but confirmed the lack of available inventory.

Polybee's growth depends on rising drone adoption, but U.S. regulatory risks to its key partner DJI threaten market access.

Polybee Plan to Address DJI Risks

Polybee acknowledges the current U.S. shortage of DJI drones and attributes it to market expectations of future restrictions. As an authorised DJI dealer, it holds priority access to stock in Singapore and Australia, enabling global shipments as part of its end-to-end service. In the event of a U.S. ban on DJI by year-end, Polybee has identified alternative hardware pathways: the Skydio Dock, which is 2–3 times more expensive, or a combination of Parrot AI drones and a third-party battery swapping system, approximately 1.5 times more expensive. The company has prior operational experience with Parrot systems and is already in discussions with U.S. manufacturers for swapping solutions.

Polybee believes that while these alternatives carry higher upfront costs, the recently enacted FAA BVLOS (Beyond Visual Line of Sight) regulations could offset them by doubling operational coverage per unit and improving efficiency through reduced downtime. This means that, under either alternative, Polybee expects operational economics to remain viable and competitive despite a potential DJI ban.

¹⁴ <https://www.aspistrategist.org.au/eyes-over-the-paddock-farm-monitoring-with-chinese-drones-is-a-national-security-risk/>

¹⁵ https://www.faa.gov/uas/advanced_operations/dispensing_chemicals

¹⁶ <https://dronelife.com/2025/06/25/dji-drones-us-consumer-market-scarcity-security/>

¹⁷ <https://dronelife.com/2024/12/08/fy-2025-ndaa-conference-text-what-happened-with-the-countering-ccp-drones-act/>

¹⁸ <https://dronelife.com/2025/06/25/dji-drones-us-consumer-market-scarcity-security/>

¹⁹ <https://www.theverge.com/report/692398/dji-drones-out-of-stock-best-buy-amazon-deny-us-exit>

Polybee has a clear hardware substitution plan that leverages both established relationships with alternative suppliers and favourable U.S. regulatory changes to preserve its economic model in the event of a DJI ban. The strategy appears operationally credible, though execution risk remains in adapting systems to non-DJI platforms.

Drone Market Share in the U.S

Rank	Manufacturer	Location	Founded	US Market Share
1	DJI	Shenzhen, China	2006	76.10%
2	Intel	Santa Clara, USA	1968	4.10%
3	Yuneec	Hong Kong, China	1999	2.60%
4	Parrot	Paris, France	1994	2.50%
5	3DR	Berkeley, USA	2009	0.60%
6	Autel Robotics	Bothell, USA	2014	0.60%
7	Skydio	Redwood City, USA	2014	0.30%
8	SenseFly	Lausanne, Switzerland	2009	0.20%
9	Kespry	Menlo Park, USA	2013	0.10%
10	AeroVironment	Simi Valley, USA	1971	0.10%

Source: Dronelife²⁰

Analysis of DJI Alternatives

The FAA's new Beyond Visual Line of Sight (BVLOS) rules double drone coverage per unit, offsetting higher US hardware costs and removing reliance on DJI. With Skydio Dock (US), Polybee can cover 200 hectares of spinach and broccoli annually, generating US 200k over two years. Despite a higher capex of US 43k, profitability is reached in 6.5 months—only slightly longer than with DJI.

With Parrot drones (France) and Hextronics' battery swapping system, the economics improve further. Covering the same 200 hectares, revenues remain US 200k over two years, but with lower capex of US 34k and payback in 5.5 months. Automated battery swapping also enables continuous operations.

Both options show that Polybee's business model is hardware-agnostic. While US-made systems cost more upfront, BVLOS rules expand service capacity, preserving margins and ensuring resilience against trade restrictions. Polybee's proprietary AI stack likely secures continuity regardless of platform choice.

²⁰ <https://dronelife.com/2021/03/05/has-the-u-s-china-trade-war-changed-djis-drone-market-share-the-latest-from-drone-industry-insights/>

Competitors

Company	No. of Crops	Yield Forecast	Stress Detection	Auto Pollination	Auto Data Acquisition	Vision-based Plant Assessment
Polybee	5	✓	✓	✓	✓	✓
Arugga AI	2	✓	X	✓	X	✓
Source.ag	3	✓	X	X	X	X
Yield Computer	2	✓	✓	X	X	✓
Orchard Robotics	2	✓	X	X	X	✓

Arugga AI

Arugga AI was founded in 2017 and is headquartered near Tel Aviv, Israel²¹. It develops autonomous ground robots for greenhouse farming, including robotic pollination, plant-lowering, pruning, pest and disease detection, and yield forecasting. It was founded by Iddo Geltner (CEO) and Eytan Heller; Geltner brings deep engineering and medical-device R&D experience, while Heller adds commercial strategy and international business development expertise. Their most recent funding round was a US 4m Series A in May 2021.²²

WIPO application number WO 2020/095290 A1 (Arugga) describes a ground-based plant treatment system comprising dedicated treatment channels for pollination.²³ The treatment channels are configured to induce vibrations at a specific frequency of 100 Hz or higher, on flowers or parts of plants either via contact or in a contactless method using pulsated air flow.

Polybee claims to differentiate by offering drone-based pollination that works both in open-field and protected environments, using aerial airflow rather than ground-based contact, enabling flexible, labour-saving pollination beyond greenhouse border.

Source.ag

Source.ag was founded in November 2020 and is based in Amsterdam, the Netherlands. It provides AI-driven software that creates digital twins of greenhouse crops, optimising resource use, yield forecasting, irrigation and crop timing.²⁴ Founders Rien Kamman (CEO) and Ernst van Bruggen (CPO) met during engineering studies and later reconnected during MBAs; they drew on their data-science background at BCG and McKinsey to build AI tools for growers. Its latest fundraising was a

²¹ <https://www.arugga.com/>

²² https://www.crunchbase.com/funding_round/arugga-ai-farming-series-a-221b7f70

²³ <https://patentimages.storage.googleapis.com/90/13/44/da16e92b832bb1/US20240090395A1.pdf>

²⁴ <https://www.agribusinessglobal.com/author/source-ag/>

Series A round of approximately US 23 m in February 2023, following an earlier US 10m seed raise in March 2022.²⁵ Polybee stands apart by combining autonomous drones with AI for both pollination and yield forecasting in both open-field and greenhouse settings, whereas Source.ag focuses on fixed greenhouse software.

Yield Computer

Yield Computer was founded in 2018 and is based in Eindhoven, Netherlands. It offers an AI-based yield-forecasting platform for soft-fruit growers—particularly strawberries—providing real-time insights into crop development to improve harvest planning and sales decisions. Polybee's differentiator lies in using drones to combine autonomous pollination with yield data collection, rather than relying on fixed cameras or software²⁶, and covering both pollination and forecasting in one system.

Orchard Robotics

Orchard Robotics was founded in 2022 by Charles Wu (a former Thiel Fellow) and is headquartered in Seattle, USA. It offers vehicle-mounted AI-powered camera systems and the Orchard OS platform to capture data on fruit count, size, colour, and growth rates, helping fruit growers optimise yield quality and supply chain planning.²⁷ Charles Wu founded it to bring precision and automation to orchard management, using applied computer vision and robotics. Its latest funding was a US 3.8m Seed round on 28 March 2024, led by General Catalyst and others. Polybee differentiates by delivering mobile, drone-based pollination and yield forecasting across field and greenhouse environments, offering active intervention rather than passive observation and data collection.

²⁵ <https://www.crunchbase.com/organization/source-ag>

²⁶ <https://www.yieldcomputer.com/>

²⁷ <https://globalaginvesting.com/orchard-robotics-gains-3-8m-to-build-robots-and-ai-to-power-precision-crop-management>

Crop Selection & Go-To-Market Strategy

Polybee believes its technology can be tailored across 25 major crop categories which satisfy the following criteria:

- High farm-gate value
- Are grown by large (50Ha), vertically-integrated farms with year-round production needs
- Crops which require pollination (1/3rd of crops),
- Crops which could benefit from yield forecasting and stress detection capabilities

Polybee enters markets where its dual value proposition—pollination + yield forecasting—creates both direct yield uplift and operational savings. It focuses on crops where a few percentage points in accuracy or yield gain equate to tens of thousands in extra revenue per hectare. It also targets crops with influential, well-networked growers who can accelerate adoption through demonstration trials.

Polybee has focussed on the following pollinated crops:

Pollinated Crops Targeted by Polybee	Farmgate value US b
Tomatoes	16
Grapes	93
Apples	92
Tea leaves	81
Oil palm fruit	68
Cucumbers and gherkins	67
Watermelons	55
Onions and shallots, dry (excluding dehydrated)	48
Chillies and peppers, green (Capsicum spp. and Pimenta spp.)	40
Eggplants (aubergines)	31
Oranges	29
Tangerines, mandarins, clementines	26
Strawberries	26
Spinach	22
Pears	20
Cauliflowers and broccoli	17
Lettuce and chicory	16
Carrots and turnips	16

Cantaloupes and other melons	15
Pineapples	14
Pumpkins, squash and gourds	10
Kiwi fruit	5
Blueberries	3

Tomatoes and blueberries sit at opposite ends of total farm-gate value, but both can be attractive for Polybee for different strategic reasons.

Tomatoes offer high per-hectare value – With ~US 53,000 per hectare, where a modest yield gain translates into significant extra revenue for growers. It's also a crop that offers production concentration – In the U.S, about 1.2 % of vegetable farms produce over half the nation's output. This means Polybee can reach a large share of the market through a few major grower relationships.

Unlike tomatoes, blueberries require insect pollination for optimal yield and fruit quality. Pollination shortfalls can cause major revenue loss, making Polybee's service directly relevant to grower pain points. Even though the total market is small (US 3 b globally), the per-hectare value is around US 25,000, meaning Polybee's service fee (US 1,000/ha) is a small percentage of revenue, yielding strong ROI.

Technology Assessment

Pollination Technology

Polybee's pollination solution uses autonomous aerial drones to deliver targeted pollination in controlled environments such as greenhouses. The system replaces or supplements manual pollination and natural pollinators (e.g., bees, bumblebees), which are often impractical or unreliable in indoor settings. The technology combines drone mobility with precision imaging and automated flight paths to locate flowers and deliver pollen efficiently.

The drones are designed to operate in confined greenhouse spaces without damaging plants. A core feature is the use of onboard vision systems to detect flowers at the correct stage for pollination, combined with a delivery mechanism that vibrates or applies pollen directly to the stigma to promote fertilisation.

This means that the system aims to increase fruit set consistency, improve yields, and reduce reliance on labour, which is increasingly scarce and costly in horticulture.

Polybee's Pollination Technology (ACP)

Polybee's solution incorporates aerodynamically controlled pollination (ACP), a proprietary contactless mechanism for self-pollination, using off-the-shelf drones which can self-fly and self-charge relying on DJI drone infrastructure.

The main benefit of ACP is precision and uniformity over crop cycles, which directly improves yield quality and marketable output. Inconsistent pollination can lead to smaller, misshapen, or unevenly ripened fruit, which reduces grade and selling price. Aerodynamically controlled pollination delivers a consistent airflow or vibration pattern targeted at each flower cluster, ensuring pollen reaches receptive surfaces without damaging delicate petals or dislodging immature fruit. This means that growers can achieve:

1. Higher fruit set rates by reaching more flowers at the optimal stage.
2. Improved fruit uniformity through even fertilisation across the crop.
3. Reduced reliance on bees in greenhouses or regions with low pollinator populations.
4. Lower labour costs compared with manual pollination, while operating at scale in large berry tunnels or indoor farms.

How ACP it Works & Effectiveness

As the drones fly over plants, the airflow downwash vibrates the flowers at suitable frequencies which releases pollen from the male parts, and it ends on the female parts within the same flower. Polybee claims that airflow downwash has been derived from an empirical formula, which is the company's intellectual property, that takes in drone form factor and propeller size, flower dimensions as the input, and computes flight plan as an output.

ACP Offers up to 3x Pollination Efficacy Compared to Natural Ventilation

Experimental results demonstrate that Polybee's Aerodynamically Controlled Pollination (ACP) technology delivers a statistically significant improvement in pollination efficacy. In controlled trials on greenhouse strawberries, ACP achieved an average of 73% pollinated stigmas compared with 23% under natural ventilation. An ANOVA test returned a p-value of less than 0.0001, indicating an extremely low likelihood that these results occurred by chance. The trials were conducted on a large sample size, adding robustness to the findings. If current industry practices deliver pollination outcomes closer to the 23% baseline, ACP offers more than a threefold improvement, representing a strong and quantifiable competitive advantage.

Commentary on Difference with Arruga

WIPO application number WO 2020/095290 A1 (Arugga) describes a ground-based plant treatment system comprising dedicated treatment channels for pollination.²⁸ The treatment channels are configured to induce vibrations at a specific frequency of 100 Hz or higher, on flowers or parts of plants either via contact or in a contactless method using pulsated air flow.

According to Polybee, Arugga's method may offer more controlled vibration transfer per flower, but at the cost of slower throughput and potential physical wear points. Polybee's ACP on the other hand, uses a non-contact approach can be faster for scanning and treating many flowers, and avoids mechanical wear and contamination risks.

Yield Forecasting Technology

Polybee has provides advanced yield forecasting technology to cater towards open-field and protected cropping systems (greenhouses). It uses AI image analysis to provide accurate 3D digital twin with centimetre level accuracy and near-real time predictions that optimise harvest planning and resource allocation (e.g water treatment).

Data collection is conducted at regular intervals with drones to ensure consistent and reliable insights. The collected data is then processed and visualized on an interactive dashboard called Phenobee, allowing users to track trends, monitor crop health, and analyse yield forecasts over time.

Provided Yield Gains of 17-30% for Spinach with Boratto Farms in Australia

Polybee's yield forecasting platform combines drone-based imaging, multispectral sensing, and proprietary AI models to deliver early, accurate, and actionable harvest projections for high-value horticulture crops. The system addresses three major profitability drivers for growers—yield, pricing, and cost—by enabling precise harvest timing, early stress detection, and improved market alignment.

The technology operates by collecting high-resolution plant imagery through autonomous drones, such as the DJI M3M multispectral platform²⁹, with typical collection frequencies of two to three times per week. Data is transferred to the cloud and processed within 12–24 hours. AI models measure key traits used by growers—leaf size for spinach, head size for broccoli—sampling over 100,000 leaves per row or detecting head size with <1 cm error.³⁰ This data is integrated with growing degree days (GDD) and micro-climate information to forecast growth rates and predict optimal harvest windows.

²⁸ <https://patentimages.storage.googleapis.com/90/13/44/da16e92b832bb1/US20240090395A1.pdf>

²⁹ <https://aq.dji.com/mavic-3-m>

³⁰ Polybee – Forecast Harvest Readiness and Yield with Drones and AI.pdf

In spinach trials with Boratto Farms in Australia, Polybee's system improved yields by up to 20% when its recommendations were followed, detecting plant stress up to five days earlier than human scouts. Case studies demonstrated yield gains of 17–30% from harvesting on the advised dates.³¹ For broccoli, the platform achieved >95% accuracy in stand counts and accurate head size forecasts up to seven days in advance, enabling better in-spec recovery rates, reduced harvest passes, and optimised labour allocation.

Polybee's web-based dashboard presents row-level forecasts, normalised difference vegetation index, growth trends, heat maps, and harvest readiness insights. A "Harvest Optimiser" module uses historical and variety-specific growth data to recommend planting and harvesting schedules aligned with supply targets. Insights can be accessed via desktop, tablet, smartphone, or even on-site displays for at-a-glance decision support.

The value proposition centres on reducing subjective decision-making, preventing yield loss from premature or delayed harvesting, and enabling evidence-backed negotiations with buyers. The platform delivers measurable operational efficiencies, such as saving 4–5 hours per week in manual scouting, and strengthens supply chain confidence by reducing inconsistencies in volume and quality. This means that adoption can translate directly into higher revenues, lower costs, and reduced waste across large-scale horticultural operations.

IP Ownership and Status Commentary

ACP Patent Status

Polybee claims that its Aerodynamically Controlled Pollination (ACP) is patent-pending in strategic markets.³² Upon review, Polybee is actively pursuing protection in key markets (Japan, Canada, Europe, Australia) but has forfeited the U.S. market for its ACP invention. The loss of U.S. coverage removes patent exclusivity in one of the largest global agriculture technology markets.

Jurisdiction	ID	Assigned	Filed	Published	Status
Japan	JP2023554240A	Yes	19/11/2021	17/12/2023	Pending
Canada	CA3199668A1	Yes	19/11/2021	27/05/2022	Pending
WIPO (PCT)	WO2022108533A1	Yes	19/11/2021	27/05/2022	Pending
Europe	EP4247146A4	Yes	19/11/2021	13/11/2024	Pending
Australia	AU2024266942A1	Yes	27/11/2024	19/12/2024	Pending
US	US20240090395A1	Yes	19/11/2021	21/03/2024	Abandoned

Source: Google Patents³³

We questioned the company why the patent in the U.S was abandoned. Polybee advised that the reason for the change was because Polybee engaged U.S. patent firm Fenwick, recommended by advisor Jorge Heraud (founder of Blue River, acquired by John Deere), due to their strong agtech expertise. Following the first office action on the original Singapore-filed application, Fenwick advised

³¹ Polybee – Forecast Harvest Readiness and Yield with Drones and AI.pdf

³² <https://polybee.co/technology>

³³ <https://patents.google.com/patent/US20240090395A1/en?q=18038264>

abandoning it and filing a continuation application to refine claims and improve the likelihood of earlier grant. Fenwick's relationship with the assigned patent examiner provides insight into preferred claim structures. The continuation maintains the original priority date while enabling claim enhancements.

Paspalis has cited the continuation applications submitted to the United States Patent and Trademark Office made on 16th of May 2025.

Yield Forecasting Protection

Polybee further claims that it has filed patent for fruit tracking and yield forecasting algorithms. However, these have not been cited. It was advised by Polybee there is a method which is patent-worthy but remains a trade secret for the time being.

Data Ownership and Potential for Monetisation

Paspalis engaged Polybee to understand the data ownership model as a potential revenue stream.

Polybee's data governance model distinguishes between raw farm data and the insights generated by its AI-driven analytics. Under current arrangements, raw data collected on-site via drones and sensors remains the property of the farmer. Polybee licenses the processed insights back to the farmer as part of its subscription service. These insights are subject to confidentiality agreements designed to protect the grower's competitive advantage and proprietary farming intelligence.

Polybee is permitted to use farm data for internal product development and R&D purposes, provided privacy and competitive positioning are maintained. This includes using anonymised datasets to:

- Train AI models across diverse crops and conditions.
- Enhance algorithms for improved forecasting accuracy and expanded feature sets.
- Validate new product features across multiple geographic and climatic zones.

At present, Polybee does not sell identifiable farm data to third parties. Potential monetisation avenues for anonymised, aggregated datasets could include industry benchmarking tools, supply chain optimisation services, and licensing to agronomic research bodies. These options would require careful balancing of farmer trust, regulatory compliance, and commercial opportunity and is being explored by the company.

NT Economic Benefit

Mangoes in the NT

The NT grows about half of Australia's mangoes each year and generates farm-gate values of over AUD 128m to the NT. NT Farmers and ABC Rural both report ~52% of national volume, with NT also supplying the first fruit of the season. This means that even small gains in yield or pack-out move national supply and NT farm income.

NT mango growers earn the highest prices in the first weeks of the national season because they face little competition. As soon as Queensland, WA, and later-season NT blocks enter the market, national supply rises quickly. This increase in supply pushes wholesale prices down, sometimes sharply, even if demand stays stable.

Historical price data from Brisbane, Sydney, and Melbourne markets shows that early NT trays can sell at a significant premium (often double or more) compared to prices seen when the market is fully supplied. Once the "flood" from later regions arrives, NT growers still selling fruit experience lower returns, even if quality remains high.

Conversations with NT Plant Industries

Polybee has held discussions with the Director of NT Plant Industries, Dr Mohammad Sohail. NT Plant Industries operates within the Northern Territory Department of Agriculture and Fisheries and serves as the Territory's primary research and development hub for plant-based industries encompassing horticulture, cropping, forestry, and pastures.

Polybee and NT Plant Industries have resulted in pursuit of collaboration opportunities with Razor Rock Farms³⁴ and Fortuna Produce³⁵, both of which have validated their interest through recent site visits, as well as NT Plant Industries research facilities for agronomic support and industry liaison capabilities.

Polybee and NT Plant Industries have identified two problem statement for NT mangoes.

1. Size-spec premiums drive economics — Export markets want fourteen-fruit trays. Fruit outside that range suffers severe price penalties or becomes unsaleable.
2. Single-pass harvesting — Because mango cultivars in commercial NT orchards ripen synchronously, growers can't pick only the optimally sized fruit and leave the rest to size up. They get one shot per block, so knowing the size distribution before harvest is critical.

Polybee's aims to resolve these issues by:

- 3D fruit tracking and point cloud reconstruction to measure fruit count and dimensions during growth.
- Allometric growth models for NT's main cultivars to predict size at harvest.
- Thermal imaging to model heat unit accumulation and link it to harvest readiness.

This means that Polybee can potentially give NT growers two commercial advantages:

³⁴ <https://www.agriculture.gov.au/about/news/leading-water-management-razor-rock-farm>

³⁵ <https://www.fortuneagri.com/about-us>

1. Maximise export-grade proportion — Harvest at the point when the highest share of fruit meets the fourteen-fruit spec.
2. Hit market windows — Align harvest maturity with premium export pricing before other growing regions flood supply and drive prices down

Potential Economic Benefit in the NT

To better understand the potential for Polybee in the NT Mango Market we modelled the potential annual revenue uplift from deploying Polybee's pollination and yield forecasting technology in the Northern Territory's mango industries. The model uses NT Farmers' reported 2023 farm-gate values — approximately AUD 128m for mangoes as the baseline.

Incremental value was calculated as:

Farm-gate value × percentage improvement in Class-1 pack-out × adoption rate.

We applied improvement scenarios of 3%, 5%, and 7% to reflect modest but realistic gains from improved pollination, fruit sizing, and harvest timing. Adoption rates were modelled at 25%, 50%, 75%, and 100% of NT production.

		NT Adoption Rate of Polybee by NT Mango Growers					
Yield Improvement		5%	10%	25%	50%	75%	100%
	1%	0.1	0.1	0.3	0.6	1.0	1.3
	2%	0.1	0.3	0.6	1.3	1.9	2.6
	3%	0.2	0.4	1.0	1.9	2.9	3.8
	4%	0.3	0.5	1.3	2.6	3.8	5.1
	5%	0.3	0.6	1.6	3.2	4.8	6.4
	6%	0.4	0.8	1.9	3.8	5.8	7.7
	7%	0.4	0.9	2.2	4.5	6.7	9.0
	8%	0.5	1.0	2.6	5.1	7.7	10.2

As can be seen, assuming 25% of NT growers adopt Polybee and an average yield increase generated of 5% could potentially generate up to AUD 1.6m in additional revenue per year for NT Growers. In a best case scenario, if Polybee's technology yield forecasting technology, is proven effective at NT scale, could deliver meaningful economic impact for two of the Territory's most important horticultural industries, particularly when leveraged to capture early-season and export premiums.

For Polybee, widespread adoption in mangoes opens up a new market in where this kind of yield forecasting technology has not been used before. It also aligns to Polybee's strategy. Mangoes is considered a high value crop, approximately AUD 20,000 per hectares in the NT where most of the local production is concentrated in the Darwin-region.

Benefit to Asian Markets

Territory mangoes are exported to key Asian markets (including Singapore). NT mangoes are attractive to export markets not only for their delicious taste but also for being counter cyclical to major northern hemisphere producers.

Melons in the NT

A similar model was used to forecast the potential revenue increases for the NT melon industry, worth about AUD 69.4m to the NT.

NT Adoption Rate of Polybee by NT Melon Growers

		5%	10%	25%	50%	75%	100%
Yield Improvement	1%	0.0	0.1	0.2	0.3	0.5	0.7
	2%	0.1	0.1	0.3	0.7	1.0	1.4
	3%	0.1	0.2	0.5	1.0	1.6	2.1
	4%	0.1	0.3	0.7	1.4	2.1	2.8
	5%	0.2	0.3	0.9	1.7	2.6	3.5
	6%	0.2	0.4	1.0	2.1	3.1	4.2
	7%	0.2	0.5	1.2	2.4	3.6	4.9
	8%	0.3	0.6	1.4	2.8	4.2	5.6

Like Mangoes, even at partial adoption and lower gains, the model demonstrates million-dollar potential, reflecting the high per-hectare value of these crops and the sensitivity of returns to small changes in marketable yield.

Anticipated Polybee Projects in the NT 2025-26

Based on engagement with NT Plant Industries, Polybee has included the following projects in its MOU with the Department.

Crop & Focus Area	NT Problem Statement	Research Hypothesis	Potential Commercial Partners Spoken With
Watermelon – Crop Stress Measurement & Yield Forecasting	Manual scouting detects fungal stress too late; uneven ripening causes inefficient harvest scheduling and profit loss.	Multispectral (680, 730, 840 nm) and thermal IR (8–14 µm) imaging via drones to detect early stress; adapt 3D fruit tracking for size forecasting and harvest optimisation.	Red Dirt Farms; NT Plant Industries research sites.
Mango – Yield Forecasting	Strict export size specs; single-pass harvests require accurate pre-harvest yield prediction.	Adapt 3D fruit tracking and point cloud reconstruction with mango-specific datasets and growth models; integrate thermal imaging to predict optimal harvest timing.	Razor Rock Farms; Fortuna Produce; NT Plant Industries research facilities.
Citrus – Ripeness Measurement	Warm climates lack reliable colour change cues; subjective assessment causes poor harvest timing.	Thermal IR drones to track accumulated heat units; develop predictive algorithms validated by acidity and peel oil content analysis.	Research partner to be confirmed.

Sales Strategy

Notable Partnerships

The company has adopted a multi-pronged sales strategy focused on channel partnerships and industry association engagement to accelerate market penetration in strategic geographies.

Strategic Channel Partnership – Bayer Crop Science

The company has a channel partnership with Bayer Crop Science, the world's largest agricultural input company.³⁶ This relationship provides privileged access to Bayer's grower customer base in key Australia–New Zealand and U.S. markets. Bayer has facilitated targeted introductions through its commercial operations team, leading to signed customers including Leaderbrand, and Qualipac. The partnership also secures priority exhibition slots at grower field day events, enabling direct presentations to Bayer's customer base. While final commercial terms are pending, the partnership has already generated measurable customer acquisition benefits.

Industry Association Engagement – Western Growers Association

The company is a member of the Western Growers Association (WGA), the largest representative body for fresh produce growers in Western USA.³⁷ WGA has facilitated direct engagement with prospective customers such as Taylor Farms, the world's largest produce grower. Through its established grower relationships, WGA provides hands-on support that reduces the sales cycle and improves conversion rates in the U.S. market.

³⁶ <https://www.crop.bayer.com.au/>

³⁷ <https://www.wga.com/>

The Polybee Sales Process

The company's sales process is designed to maximise customer value while reducing churn risk, which is critical when targeting large, vertically integrated produce companies. The current sales strategy seems to be a well-aligned to high-value customer acquisition in large-scale horticulture. The staged approach mitigates adoption barriers and builds strong commercial traction, but future growth will depend on building a scalable sales team that can replicate this relationship-driven model in multiple geographies.

Sales Process Stage	Description	Key Metrics / Commercial Detail
Screening	Target customers are vertically integrated produce companies producing more than 100 hectares annually. This ensures potential annual account size of at least US 50,000.	30 paying customers targeted in next two years (currently engaging with 12). Median account potential: US 250k
Qualification	Senior sales team (VP Commercial Operations and CEO) engage with VPs/CXOs and Heads of Operations to present value proposition and understand priorities. If aligned, a first-stage deployment is proposed.	2–3 conversations on average to secure first-stage commitment.
Stage-1 Deployment	Low-risk pilot covering 10–20% of cropped area over 3 months. Priced fairly to allow customers to test ROI before scaling.	Enhances retention and de-risks large-scale operations.
Stage-2 Deployment	Full-scale rollout priced at US 500 per hectare per crop cycle. Scope includes drone units, accessories (portable power supply, Starlink) and location-specific adjustments. Technical specialists handle on-site setup and onboarding.	In Australia, product goes live within 7 days from contracting.

Business Model Under Stage 2

Polybee charges *per hectare, per crop cycle* based on proven value generate in per-crop trials with customers.

This model claims to work because:

1. It directly links to the grower's revenue potential for that specific crop and cycle.
2. Crops with higher farm-gate value per hectare can better absorb the cost of Polybee's service while still giving the grower a strong return on investment.
3. It standardises comparisons between crops with different growth durations and land productivity.

The per-hectare, per-crop-cycle pricing model is argued as a strength. It ensures that customers realise measurable returns within a single crop cycle, directly linking spend to value creation. Pricing was validated through case studies with early customers to maintain significant ROI multiples on the fixed rate. For example, in documented trials with Boratto Farms, the technology delivered net profit improvements of over US 3k per hectare in spinach crops, equating to up to a 6x return on the US 500/Ha price point. This short payback period and high ROI potential are compelling for customers operating in tight-margin fresh produce markets.

The combination of rapid value demonstration, minimal technical burden on the customer, and strong economic returns creates a robust commercial proposition. This means that the model is well-positioned to support both customer acquisition and retention, particularly among high-value accounts. The primary execution risk lies in ensuring the operational capacity to deliver seamless onboarding at scale while maintaining service quality across multiple geographies.

Unit Economics & Gross Margins

Boratto Farms Example Showing Potential for 65% Gross Profit Margins

Polybee has provided a real-world example outlines the economics of a 100-hectare open-field spinach and broccoli operation at Boratto Farms, a Stage-2 customer. The farm is serviced by a single autonomous drone system with dock, embedded computer, and router.

Under the per-hectare, per-crop-cycle pricing model, the account generates US 50,000 in annual revenue, or US 100,000 over two years. Polybee's capital expenditure for the deployment is US 16,184, covering all hardware components. Ongoing operating expenses, primarily internet connectivity and cloud service fees, average US 783 per month (US 18,792 over two years).

With average monthly revenue of US 4,166, the payback period for the initial investment is approximately five months, after which the engagement yields a strong gross margin for the remainder of the term. This rapid payback underscores the profitability potential of high-hectare accounts.

Item	US	Notes
Revenue	100,000	US 500/Ha × 100 Ha × 2 years
Capital Expenditure (Capex)	16,184	Drone, dock, embedded computer, router
Operating Expenditure (Opex)	18,792	Internet and cloud service fees (US 783/month × 24 months)
Total Costs	34,976	Capex + Opex
Two-Year Profit	65,024	Revenue – Total Costs
Profit Margin	65%	Profit ÷ Revenue × 100

Financial Commentary

Historical Summary

A general historical and financial summary is provided based on what has been provided the company:

SGD	Historical/Actual			Forecast		
	2022	2023	2024	2025	2026	2027
Revenues	322,561	232,860	86,572	236,500	2,202,600	5,329,600
Gross Profit	304,163	142,136	(26,573)	131,187	981,075	2,162,328
Other income	370,406	161,988	647,228	-	-	-
Operating expense	(1,237,798)	(1,159,944)	(1,281,895)	(2,013,111)	(3,318,293)	(3,965,338)
NPAT	(563,229)	(855,820)	(574,666)	(1,294,815)	(1,859,257)	(662,395)
Cash EOP	2,453,702	1,796,641	1,163,873	-	-	-
Net change in cash	2,453,703	(622,155)	(634,619)	-	-	-
Salary expense		(734,589)	(902,934)	(1,622,638)	(2,961,523)	(3,582,340)

Forecast is based on the company's 3-way forecast. This Forecast assumes the year end is December 31st per the Polybee's ACRA profile. Note that 2025-2026 pipeline value of US 2.1m is considerably less than the 2025 financial year projections.

Revenue decline explanation

Revenue declined from SG 322k in 2022 to SG 86k in 2024. In 2022 and 2023, a significant proportion of the company's revenue came from pilot projects with major agricultural input companies, including Bayer and Syngenta. These projects applied the company's yield forecasting technology to digital phenotyping for internal product development. The pilots achieved their technical objectives; however, anticipated commercial scale-up in 2024 did not materialise. This has been alleged by management due to a sector-wide downturn in the seed industry, with companies such as Bayer reporting negative cash flow, implementing workforce reductions of approximately 25%, and substantially cutting budgets for new product initiatives.

In response, the company pivoted to a direct-to-grower model. While the underlying technology was already proven, this shift required additional product development to deliver demonstrable and repeatable return on investment for growers. The company dedicated 2024 to building these capabilities and aligning its offering to secure product-market fit. Management reports that the current customer pipeline is sufficient to achieve this milestone. Investors were informed of the strategic shift and supported the decision, acknowledging that 2024 revenue projections would be materially reduced as a result.

Other revenue is all R&D grant tranches from Hort Innovation and Enterprise Singapore.

Expenses and Balance Sheet Comments

Salaries as a percentage of total operating expenses for 2023 and 2024 were 63% and 70% respectively. Travel expenses also appear to make up a high proportion of expenses, and SG 117,827 in 2023 and SG 125,748 in 2024. Travel is likely linked to the research and development setup of paid trials under Stage 1 and 2 of its sales process, however as revenue has declined over but 2023-24, it suggests that this travel is not translating to revenue. Travel expenses should be monitored carefully given the need expand overseas in Australia and the United States and local hiring strategies in these countries which could otherwise manage this expense over time. Research and development is low, SG 1,253 in 2023 and SG 73,239 for 2024. This is likely due to the use of

off-the-shelf drone technology and reliance on specialised engineering staff to complete yield forecasting development as a salary expense item.

The cash conversion ratio has generated been close to 100%. From 2022-24, net cash flow conversion has been 123%, 72% and 102% respectively. These fluctuations highlight that operating cash performance has been sensitive to the timing of supplier payments, customer receipts, and inventory movements, which may not be sustainable or predictable – typical of a seed stage startup. The movement in Trade and Other Payables shifted from a cash outflow of SG 221,389 in 2022 to a cash inflow of SG 231,752 in 2023, representing a year-on-year swing of SG 453,141. This improvement in operating cash flow is not the result of higher profitability, but of changes in working capital. The positive movement in 2023 likely reflects delayed payments to suppliers, increased purchasing activity toward year-end, extended credit terms, or the inclusion of new accrued liabilities. In contrast, the negative movement in 2022 indicates that the company paid down a larger portion of outstanding liabilities during that year.

Revenue Pipeline and 2026 Year

Polybee forecasts that it will generate US 2.1m over 2025-2026 most of these are from Stage 1 trial which occur over a three month period. The cost charged to customer is US 500 per Ha.

Crop	Customer	Country	Ha	US	Engagement Stage	Notes
Spinach	Boratto Farms	AU	100	50,000	Stage 2	
	Leaderbrand	NZ	150	75,000	Stage 1	>50% market share in spinach and broccoli in New Zealand
	Harvest Moon	AU	90	45,000	Stage 1	Top 5 spinach and broccoli grower in Australia
	Dicky Bill	AU	400	200,000	Discussions	Top 5 spinach grower in Australia
	Bulmer Farms	AU	680	340,000	Discussions	Top 5 spinach grower in Australia
	Schreurs Farms	AU	320	160,000	Qualified lead	Top 5 spinach grower in Australia
Broccoli	Leaderbrand	NZ	750	375,000	Stage 1	>50% market share in spinach and broccoli in New Zealand
	Harvest Moon	AU	200	100,000	Stage 1	Top 5 spinach and broccoli grower in Australia
	Qualipac	AU	300	150,000	Stage 1	
	Simplot	AU	450	225,000	Discussions	
	T&A Produce	USA	800	400,000	Stage 1	Top 3 fresh produce grower in all of USA
Total Pipeline 2025-26				2,120,000		

Cited Contracts & Commentary on the 2025-26 Revenue Forecasts

The data room contains six contracts, which support Stage 1 and Stage 2 adoption as part of the 2025-2026 pipeline:

Partner	Amount	Duration	Stage	Crop(s)	Date
Tanimura & Antle (T&A)	US 8.5k	3 months	Stage 1	Broccoli	23 Apr 2025
Qualipac	AUD 7.5k	3 months	Stage 1	Broccoli yield forecasting	3 May 2025
Harvest Moon	AUD 15k	3 months	Stage 1	Broccoli & spinach forecasting	25 Mar 2025
Boratto Farms	AUD 76k	—	Stage 2	Spinach	2 May 2025
Perfection Fresh	AUD 15k	4 months	Stage 1	Glasshouse tomatoes (pollination)	18 Jul 2025
Powerhouse Producer	AUD 35k	3 months	Stage 1	Pollination (unspecified crop)	25 Jun 2025

The contracts reviewed highlight a clear gap between Polybee's current contracted revenue and its projected pipeline. While the company cites a US 2.1m pipeline, this figure reflects potential rather than secured revenue. Most customer relationships remain in Stage 1 trials or early engagements, with current contracts showing limited coverage of farm area at a much lower per-hectare value.

This means that near-term revenue will depend on how quickly trials are initiated and how effectively they are converted into larger Stage 2 rollouts. If customers validate return on investment and move swiftly to scale, revenues could begin to track towards the pipeline within the forecast period. However, if trial results are mixed or conversion cycles take longer than expected, 2025–2026 revenue may fall short of the implied run-rate. The largest accounts, particularly those closest to Stage 2 discussions, will play a decisive role in bridging the gap between early-stage contract values and forecast potential.

Historical and Future Cash Burn & Runway

Bank Balance and Runway

Proof of Balance	As at 31 July	Currency	SGD	Source/Note
Wise SG	84,262	SGD	84,262	Bank statements
DBS	286,198	US	372,058	Bank statements
DBS	12,985	SGD	12,985	Bank statements
Wise Australia	28,713	AUD	25,842	Bank statements
Current Cash Balance			495,146	
Cash at June 30 2025		SGD	631,458	Group BS Consolidated Q2
Cash at March 30 2025		SGD	921,878	Group BS Consolidated Q1
Change over 3 months		SGD	-290,420	
Monthly Burn		SGD	(96,807)	
Estimated Runway			5.1 months	

Forecast Burn Commentary

Based on the 3-way forecast model provided, we expected cash burn to be approximately US 200k per month, the vast majority is made up of salary expenditure increases due to hiring an additional 9 roles.

Assuming *no revenue*, Polybee hires an additional 9 roles by March 2026, and their financial model is corrected to show income tax credits not payments as a loss-making company, cash burn on a Seed Round of US 2.0m in September is expected to be **13-15 months**.

This is not an unreasonable estimate given high expectation that Stage 1 trial customers will convert to Stage 2 customers. However, given the high proportion of salary expenses to overall cash burn, if hires are made slower than expected in line with sustainable revenue growth over the period, then Polybee's runway could most likely be extended to **15-18 months**.

Series A Expectations, US 10m at 40-50m pre-money

Discussions with management suggest Series A capital raise in Q3–Q4 2026, approximately 12–15 months from the current period. At that stage, the company projects US 2m in annual recurring revenue (ARR) with a clear path to cash flow breakeven, demonstrating the viability of its capital-efficient growth model. The qualified sales pipeline is expected to exceed US 10m, supporting management's assertion of 3–5× ARR growth over the subsequent 12 months. Based on observed pipeline conversion rates with current customers, this is expected to translate into US 6–9m in next-twelve-months revenue by the second half of 2026.

Management's target pre-money valuation range for the Series A is US 40–50m. This reflects a 5–7× forward-revenue multiple, consistent with market comparables for companies with similar unit economics and growth trajectories. The planned raise size is approximately US 10m. Proceeds are intended for three primary growth drivers: geographic expansion into Europe and Latin America (where early opportunities have been identified), accelerated product development focused on new crop coverage, and scaling operational infrastructure to support anticipated customer growth. Management positions the Series A as an attractive entry point for growth investors seeking exposure to the emerging market for physical AI agents in agriculture.

Series A Expectations Commentary

Management's estimates of Series A slightly unrealistic considering Series A benchmarks. For example, Preqin estimates median Series A valuation for Global Agtech companies to be around US 30m. For example, Agurra, a Polybee competitors raised a Series A US 4m Series A in May 2021.

Average Pre-Money for Global Ag-tech Companies			
Year	Series A Post-Money US m	Average Equity Sold in Series A	Implied Series A Pre-Money, US m
2025	37.3	23.6%	28.50
2024	44	22.9%	33.92
2023	38.6	23.1%	29.68
2022	54.3	26.3%	40.02
2021	46.5	22.0%	36.27
2020	26.5	22.3%	20.59
2019	30.2	22.7%	23.34
2018	35.7	23.7%	27.24
2017	62.7	23.2%	48.15
2016	28.3	23.9%	21.54
2015	10.8	24.0%	8.21

Source: Preqin

Exit Activity

Comparable Exits

Across the five most relevant precision agriculture exits in the last five years, all involving companies founded within the past decade, transaction sizes ranged from US 60m to US 300m. The median deal size was US 153m, and the average was approximately US 175 m.

Company	Founded	Acquired by	Transaction size	Year	Relevance to Polybee
Root AI	2018	AppHarvest	US 60m (US 10m cash + shares) ³⁸	2021	Greenhouse robotics and AI for tomatoes; data for yield forecasting and harvest readiness in controlled horticulture.
&ever (vertical farming)	2015	Kalera	US 153m ³⁹	2021	Automated indoor leafy-greens production; controlled-environment data and operations relevant to stress detection and yield optimisation.
Augmenta	2016	CNH Industrial (Raven)	US 110m ⁴⁰	2023	On-machine AI vision and multispectral sensing for real-time crop monitoring and variable-rate actions; aligns with stress detection and yield improvement.
Bear Flag Robotics	2017	John Deere	US 250m ⁴¹	2021	Field autonomy and robotics; strengthens the case for automation in specialty crops and supports drone/robot-enabled horticulture workflows.
Prospera	2014	Valmont Industries	US 300m ⁴²	2021	AI-driven crop analytics using computer vision for plant health and stress detection; directly relevant to Polybee's yield forecasting and stress monitoring.

Based on this data, there seems to be some potential for 10x returns by strategic acquirers, particularly given Polybee's pre-money valuation of US 15m. However, it likely requires strong capital allocation and execution on proven ROI in high-value crops and deep technology development.

³⁸ <https://agfundernews.com/appharvest-acquires-root-ai-to-add-ai-powered-harvesting-robots-to-its-workforce>

³⁹ <https://agfundernews.com/kalera-agrees-153m-deal-for-ever-in-vertical-farming-ma>

⁴⁰ <https://finance.yahoo.com/news/cnh-industrial-buys-machine-vision-141919480.html>

⁴¹ <https://www.farmautomationtoday.com/news/autonomous-vehicles/autonomous-technology-start-up-bear-flag-robotics-acquired-by-john-deere.html>

⁴² <https://agfundernews.com/prospera-valmont-acquires-crop-analytics-startup-for-300m>

Further, including Jorge as an advisory board member provides Polybee with direct guidance from a founder who has successfully navigated one of the few notable exits in precision agtech over the past decade. As CEO and Co-founder of Blue River Technology from 2011 to 2024, Jorge built advanced computer vision and machine learning platforms that transformed weed control through the company's "see and spray" technology. This innovation reduced herbicide use by up to 90% while delivering strong environmental and commercial outcomes, leading to the company's acquisition by Deere & Company in 2017 for US 305m.

Path to Exit

Michael Lee, Managing Director of Syngenta Group Ventures, a CVC of Syngenta Group one of the world's largest agribusinesses has recently mentioned several trends driving agtech exits:

Trade sales to the ag industry (OEMs and agchem) have been the primary exit mechanism for the majority of agtech companies for almost 20 years and are likely to remain so.⁴³

Acquirers in agtech are increasingly taking a "try before you buy" approach, often through collaborations or minority investments by corporate venture capital (CVC) units.⁴⁴ This means that partnerships with OEMs or agchem corporates, or securing a CVC investor, are important precursors to a likely trade sale. It is recommended that Polybee re-engage Bayer and Syngenta, who participated in technical pilots, to maintain high engagement. As noted in "Revenue decline explanation", in 2022 and 2023, a significant proportion of Polybee's revenue came from pilot projects with these companies.

In addition, capital efficiency has been mentioned as a key determinant of exit pathways. Historical data shows that successful exits to agtech or OEM buyers occur when companies raise less than US 50m in total funding (around US 35m in the last round and US 15m across earlier rounds). According to Michael Lee, raising beyond this level reduces the pool of acquirers and requires greater conviction in future growth. Series A expectations and beyond should be modest for Polybee.

⁴³ <https://www.agtechnavigator.com/Article/2025/06/25/the-whys-of-agtech-exits-the-empirical-evidence/>

⁴⁴ <https://www.agtechnavigator.com/Article/2025/06/25/the-whys-of-agtech-exits-the-empirical-evidence/>

Returns Analysis / Valuation Cross-Check

Some cap table analysis of two cheque sizes of US 1m and US 2 m, across worst-case, base-case, and best-case outcomes. These cheque sizes roughly approximate Paspalis and SEEDS potential position post-investment.

The results show a strong venture-style return profile with asymmetry between downside and upside outcomes. In the worst-case scenario, both cheque sizes deliver a MOIC above 2x, providing some downside protection. In the base case, the return is approximately 8x, which is attractive by venture standards. The best-case scales to 20–25x MOIC, with the larger cheque producing exit proceeds exceeding US 50 m. This suggests the investment offers limited downside relative to significant upside potential, with outcomes highly sensitive to revenue growth and dilution dynamics.

Projected ARR 2031	20,000,000	US
EV/Revenue	9.5 x	
Est. Exit Value	189,360,000	US
Target MOIC	10.0 x	
Est. Post-Money Today	18,936,000	US

Industry	EV/Sales	Weighting	Adj EV/Sales
Farming/Agriculture	1.2 x	20%	0.2 x
Software (Applications)	11.5 x	80%	9.2 x
Blended Multiple			9.5 x

Source: Damodaran, Useful Data Sets

Cases	Worst Case	Base Case	Best Case
Invested	1,000,000	1,000,000	1,000,000
Ownership at Exit	3.00%	4.00%	5.00%
Projected ARR 2031	15,000,000	20,000,000	35,000,000
Blended EV/Revenue Multiple	6.0 x	10.0 x	13.0 x
Exit Value	90,000,000	200,000,000	455,000,000
Stake value at exit	2,700,000	8,000,000	22,750,000
Projected MOIC	2.7 x	8.0 x	22.8 x
Invested	2,000,000	2,000,000	2,000,000
Ownership at Exit	6.00%	8.00%	11.00%
Projected ARR 2031	15,000,000	20,000,000	35,000,000
Blended EV/Revenue Multiple	5.0 x	10.0 x	13.0 x
Exit Value	75,000,000	200,000,000	455,000,000
Stake value at exit	4,500,000	16,000,000	50,050,000
Projected MOIC	2.3 x	8.0 x	25.0 x

Seed Round Key Risks

Conversion Risk from Stage 1 Trials

Polybee's revenue pipeline relies heavily on the successful conversion of Stage 1 trials into longer-term Stage 2 contracts. At present, only three Stage 1 trials have been signed, with most other accounts still at the Stage 1 or "Discussions" phase. The forecasted annual run rate of US 2.1m is contingent on high conversion rates. If the transition from Stage 1 to Stage 2 is slower or less frequent than anticipated, actual revenue in 2025–2026 could fall well short of projections. This means that the company's near-term growth trajectory remains unproven.

Cash Runway Risk

Scaling the team before securing meaningful Stage 1-to-Stage 2 conversions could materially shorten Polybee's cash runway. If the company increases hiring in line with its growth plan but fails to meet projected revenue milestones, it may need to raise additional capital earlier than planned. This could occur in a less favourable fundraising environment, leading to dilution or a constrained ability to execute on its strategy.

DJI Drone Dependency Risk

Polybee's hardware offering depends on DJI drones for deployment. Any disruption to DJI's supply chain, regulatory restrictions on DJI imports in key markets, or strategic changes by DJI that affect third-party integrations could delay deployments or increase costs. Given that DJI has faced geopolitical and regulatory scrutiny in some jurisdictions, such dependency could introduce operational and market-entry risks in certain regions.

Seed Round Use of Funds

The company has set out four clear milestones to be achieved with the proceeds of this seed round.

Revenue Growth to US 2m ARR

The immediate commercial goal is to reach US 2m ARR within 12 months. This will be achieved by scaling current customers from small-scale pilots to full-farm deployments. The company already has commitments from major accounts and a US 4m pipeline identified. The funds will be used to build out engineering capacity, ensuring their software platform can process high volumes of image data, deliver AI insights reliably, and incorporate customer feedback.

Ensuring 100% Customer Retention

Retention is a core focus. The company will strengthen its customer success function with hires in key hubs such as Victoria and California. This will ensure smooth onboarding and verifiable ROI for their largest customers, including Perfection Fresh, Harvest Moon, and T&A Produce. Maintaining these anchor accounts will be critical to validating the product and underpinning revenue growth.

Product Expansion into New Crops

Beyond their current focus crops, the team plans to extend their platform to at least three additional crops, starting with lettuce and blueberry. The underlying AI and computer vision systems already support this, requiring only limited adjustments. Expansion will broaden the addressable market and increase customer stickiness.

Pipeline Expansion to US 10M+

Finally, the company aims to grow its pipeline from US 4m to over US 10m. This will involve deeper penetration in existing markets and early entry into Europe and LATAM, where qualified leads already exist. The company will invest in marketing campaigns, trade shows, case studies, and regional sales hires to secure marquee accounts and position itself strongly for a Series A raise.

New Hires

Polybee has provided a high-level overview of new hires expected over 2025-2026 period:

Role	Description	Type	Area	SG Month
FDE-1	Operations specialist 1 - VIC,TAS,SA,NZ	Operations	Australia	7,500
BDE-1	Customer Success Manager 1 - VIC,TAS,SA,NZ	Operations	Australia	10,000
CV-OF	Sr. CV Engineer - OF	Engineering	Singapore	8,333
CV-OF	Sr. CV Engineer - OF	Engineering	Singapore	8,333
CV-OF	Sr. CV Engineer - OF	Engineering	Singapore	8,333
CV-OF	Sr. CV Engineer - OF	Engineering	Singapore	8,333
OPS-2	Operations specialist 2 - Salinas + Yuma	Operations	USA	10,000
OPS-3	Operations specialist 3 - NT, QLD	Operations	Australia	12,500
CSM-2	Customer Success Manager 3 - NT, QLD	Operations	Australia	10,000
CSM-4	Customer Success Manager 4 - Central Valley	Operations	USA	12,500
DevOps-1	DevOps Engineer	Engineering	Singapore	10,000
				105,833

Seed Round Offer Terms

Term	Detail
Target round size	US 2.5 m
Pre-money valuation	US 15 m
Security type	Seed Preference Shares (SPS)
Founder vesting	6 years with 1-year cliff; vesting start date 28 Sept 2022
Transfers	All transfers subject to right of first refusal; Founders need Lead Investor approval before transfer
Trustee liability	Per Paspalis standard phrasing
ESOP	Unchanged – 250,000 shares (14.59% fully diluted, post-closing)
Ranking of SPS	Rank ahead of Ordinary Shares in all respects
Liquidation preference	Non-participating; higher of (a) subscription price + unpaid declared dividends, or (b) pro-rata of net proceeds
Board seat	Lead Investor (Paspalis) may appoint one Director/Observer
Tag-along and drag-along	Customary; tag-along for SPS holders if Founder sells; drag-along if ≥75% shareholders approve sale of company

Term	Detail
Information rights	Quarterly management accounts (21 days), half-year unaudited (60 days), annual unaudited (90 days), annual audited (150 days), annual budget/business plan (30 days after year-end)
Anti-dilution provisions	Broad-based weighted average adjustment
Reserved matters	Applicable to Paspalis in addition to customary Lead Investor rights: 1. Any amendment to the Constitution of the Company. 2. Changes to the share capital, including the issue of any new shares or other securities, or granting any options, warrants or convertible securities, except under the approved ESOS. 3. Alteration of rights attaching to any class of shares. 4. Redemption, repurchase, or repayment of share capital. 5. Declaration or payment of dividends or other distributions. 6. Acquisition or disposal of assets exceeding an agreed monetary threshold. 7. Borrowing or lending above agreed limits, including guarantees and indemnities. 8. Entering into related party transactions not on arm's length terms. 9. Change of business, including starting or discontinuing any material business line. 10. Incorporation or acquisition of subsidiaries, or winding-up subsidiaries. 11. Changes to the Board composition or appointment/removal of key executives. 12. Material changes to employee share schemes or incentive arrangements. 13. Commencement or settlement of legal proceedings above an agreed threshold. 14. Voluntary liquidation, winding-up, or dissolution of the Company.

Pro Forma Capitalisation Table

Paspalis is expected to hold 5.73% post-investment, assuming the following items:

- Pre-money valuation of US 15m
- No increase to the existing ESOP (advised by Polybee management)
- Paspalis investment of US 1m, matched by SEEDS for US 1m
- Elev8 participation for US 250k (advised by Elev8)
- Jorge Heraud (advisor) participation for US 50k (advised by Polybee management)
- Angel participation for US 115k (advised by Polybee management, pre-money of 15m)
- Conversion of existing SAFEs held by SEEDS, Elev8 and Graduate Investment (NUS)

		Pre-Investment		Post-Investment	
Entity	Share Type	Shares	Fully-Diluted Ownership (%)	Shares	Fully-Diluted Ownership (%)
Siddharth Jadhav	ORD	1,000,000	80.00%	1,000,000	58.37%
SEEDS Capital Pte. Ltd.	SPS	-	0.00%	213,528	12.46%
Elevate Capital VCC	SPS	-	0.00%	116,581	6.81%
Paspalis Capital Pty Ltd	SPS	-	0.00%	98,174	5.73%
Angels	SPS	-	0.00%	19,635	1.15%
Graduate Investment Pte. Ltd.	SPS	-	0.00%	9,081	0.53%
Jorge Heraud	SPS	-	0.00%	6,136	0.36%
Employee Share Option Pool ⁴⁵	ORD	250,000	20.00%	250,000	14.59%
Total		1,250,000	100.00%	1,713,135	100.0%

The existing SAFE notes will convert into a lower price per share based on a post-money valuation cap of US 15 m or a 20% discount to the existing pre-money. Given the proposed pre-money valuation is set at US 15m, the SAFEs will convert at a 20% discount.

⁴⁵ Includes allocated and unallocated ESOP

ESOP Review

We reviewed:

- Notice of Resolution filing with ACRA dated 27th July 2023 to verify the size of the allocated/unallocated ESOP share pool.
- Polybee's ESOP filing system to verify the total number of ESOP shares that have been granted to individuals (including vested, unvested components) as of August 18th 2025.
- The existing rules of the ESOP of Polybee
- Additional information provided at our request to (1) verify the proportion of unvested ESOP due to good-leavers to true-up the allocated/unallocated portion of ESOP, and (2) there is no planned increase to unallocated ESOP in this current fundraise.

We summarise the company's ESOP position with no adverse findings:

- The board resolved to allot and issue up to 250,000 ESOP ordinary shares, representing 20% of the fully diluted shares as of 27th July 2023. This is consistent with our understanding of the ESOP size.
- That of the 250,000 ESOP shares, 184,070 is allocated (including both vested and unvested components) and the remaining 65,930 ordinary shares are unallocated.
- Of those employees, directors, and contractors that were granted options which later ceased to be before the grants were fully vested, the unvested components were forfeited pursuant to the good-leaver provisions under the company's ESOP rules.
- The unallocated shares represent 3.9% of the fully diluted share capital, and there is no intention to increase the unallocated ESOP at this point in time.